Chapter 1 Heating processes

Section 1.1 Heat and temperature

Worked example: Try yourself 1.1.1

CALCULATING THE CHANGE IN INTERNAL ENERGY

A student places a heating element and a paddle-wheel apparatus in an insulated container of water. She calculates that the heating element transfers 2530 J of heat energy to the water and the paddle does 240 J of work on the water. Calculate the change in internal energy of the water.

Thinking	Working
Heat is added to the system, so Q is positive. Work is done on the system, so W is positive.	$\Delta U = Q + W$ $= (+2530) + (+240)$
Note that the units are J, so express the final answer in J.	$\Delta U = 2770\,\text{J}$

Section 1.1 Review

KEY QUESTIONS SOLUTIONS

1 C. The kinetic particle theory states that the particles in all substances (regardless of their state) are in constant motion.

2
- The chicken and the air in the oven are not in thermal equilibrium.
- Thermal energy flows from the hot air into the chicken.
- The chicken and the air in the oven are in thermal equilibrium.

3 C and D. Negative kelvin values and Celsius values below –273 are not possible because temperatures below absolute zero are not possible.

4 The temperature of the gas is just above absolute zero so the particles have very little energy.

5 **a** K = °C + 273
= 30 + 273
= 303 K

b °C = K – 273
= 375 – 273
= 102°C

6 300 K is 27°C. Higher temperatures mean molecules have greater average kinetic energy. So the average kinetic energy of the hydrogen particles in tank B is greater than the average kinetic energy of the hydrogen particles in tank A.

7 absolute zero, 10 K, –180°C, 100 K, freezing point of water

8 $\Delta U = Q + W$
$= -20 + -50$
$= -70\,\text{kJ}$

9 $\Delta U = Q + W$
$= 75 + 150$
$= 225\,\text{J}$

10 $\Delta U = Q + W$
$250 = -300 + W$
$W = 550\,\text{J}$
The scientist does 550 J of work on the sodium.

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Section 1.2 Specific heat capacity

Worked example: Try yourself 1.2.1

CALCULATIONS USING SPECIFIC HEAT CAPACITY

A bath contains 75 L of water. Initially the water is at 50°C. Calculate the amount of energy that must be transferred from the water to cool the bath to 30°C.

Thinking	Working
Calculate the mass of water. 1 L of water = 1 kg	Volume = 75 L So mass of water = 75 kg
ΔT = final temperature – initial temperature	$\Delta T = 30 - 50 = -20$°C
From Table 1.2.1, $c_{water} = 4180\ \text{J kg}^{-1}\ \text{K}^{-1}$. Use the equation $Q = mc\Delta T$.	$Q = mc\Delta T$ $= 75 \times 4180 \times 20$ $= 6\,270\,000$ $= 6.27 \times 10^6$ J transferred from the water

Worked example: Try yourself 1.2.2

COMPARING SPECIFIC HEAT CAPACITIES

What is the ratio of the specific heat capacity of liquid water to that of steam?

Thinking	Working
Table 1.2.1 has the specific heat capacities of water in different states.	$c_{water} = 4180\ \text{J kg}^{-1}\ \text{K}^{-1}$ $c_{steam} = 2000\ \text{J kg}^{-1}\ \text{K}^{-1}$
Divide the specific heat of water by the specific heat of steam.	Ratio $= \frac{c_{water}}{c_{steam}}$ $= \frac{4180}{2000}$
Note that ratios have no units since the unit of each quantity is the same and cancels out.	Ratio ≈ 2.1

Section 1.2 Review

KEY QUESTIONS SOLUTIONS

1 Water requires more energy per degree Celsius heated because the specific heat capacity of water is much greater than that of aluminium.

2 All other variables being the same, as aluminium has the highest value for specific heat capacity, it will contain the most thermal energy.

3 100 mL of water has a mass of 0.1 kg.

$Q = mc\Delta T$

$= 0.1 \times 4180 \times (20 - 15)$

$= 2090$ J

4 150 mL of water has a mass of 0.15 kg.

$Q = mc\Delta T$

$= 0.15 \times 4180 \times (50 - 10)$

$= 25\,080$ J or 25.1 kJ

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5 Remember that both ΔT and mass are proportional to energy.

Use the relationship $Q = mc\Delta T$.

If $x = 10c$ then $20c = 2x$ J.

6 The ratio of the temperature rise is equal to the inverse ratio of the specific heat capacities as $\Delta T = \frac{1}{c}$

Ratio of temperature rise $= \frac{4180}{900} = 4.64$

The temperature of the aluminium is 4.64 times that of the water.

7 B. Different states will have different specific heat capacities.

8 If 4.0 kJ of energy is required to raise the temperature of 1.0 kg of paraffin by 2.0°C, then 2.0 kJ of energy is required to raise the temperature of 1.0 kg of paraffin by 5.0°C.

So to raise the temperature of 5.0 kg, you will need five times as much energy, i.e. 5 × 2.0 kJ = 10 kJ to raise the temperature of 5.0 kg of paraffin by 1.0°C.

Mathematically: Calculate c for paraffin, so $c = \frac{Q}{m\Delta T} = \frac{4000}{1 \times 2} = 2000\,\text{J kg}^{-1}\,\text{K}^{-1}$

Then for 5.0 kg, $Q = mc\,\Delta T = 5.0 \times 2090 \times 1.0 = 10$ kJ

9 $Q = mc\,\Delta T$

$10\,500 = 0.25 \times 4180 \times (T - 20)$

$10 = T - 20$

$T = 30°\text{C}$

Final temperature = 30°C

10 $Q = mc\,\Delta T$

$-13\,200 = m \times 440 \times -30$

$m = -13\,200 \div -13\,200$

$= 1$ kg

Section 1.3 Latent heat

Worked example: Try yourself 1.3.1

LATENT HEAT OF FUSION

How much energy must be removed from 5.5 kg of liquid lead at 327°C to produce a block of solid lead at 327°C? Express your answer in kJ.	
Thinking	**Working**
Cooling from liquid to solid involves the latent heat of fusion, where the energy is removed from the lead. Use Table 1.3.1 to find the latent heat of fusion for lead.	$L_{fusion} = 0.25 \times 10^5\,\text{J kg}^{-1}$
Use the equation: $Q = mL_{fusion}$	$Q = mL_{fusion}$ $= 5.5 \times 0.25 \times 10^5$ $= 1.4 \times 10^5$ J
Convert to kJ.	$Q = 1.4 \times 10^2$ kJ

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Worked example: Try yourself 1.3.2

CHANGE IN TEMPERATURE AND STATE

3 L of water is heated from a fridge temperature of 4°C to its boiling point at 100°C. It is boiled at this temperature until it is completely evaporated. How much energy in total is required to raise the temperature and boil the water?	
Thinking	**Working**
Calculate the mass of water involved.	3 L of water = 3 kg
Find the specific heat capacity of water from Table 1.2.1.	$c = 4180\ \text{J kg}^{-1}\ \text{K}^{-1}$
Use the equation $Q = mc\Delta T$ to calculate the heat energy required to change the temperature of water from 4°C to 100°C.	$Q = mc\Delta T$ $= 3 \times 4180 \times (100 - 4)$ $= 1\ 203\ 800\ \text{J}$
Find the specific latent heat of vaporisation of water.	$L_{vapour} = 22.5 \times 10^5\ \text{J kg}^{-1}$
Use the equation $Q = mL_{vapour}$ to calculate the latent heat required to boil water.	$Q = mL_{vapour}$ $= 3 \times 22.5 \times 10^5$ $= 6\ 750\ 000\ \text{J}$
Find the total energy required to raise the temperature and change the state of the water.	Total $Q = 1\ 203\ 800 + 6\ 750\ 000$ $= 8 \times 10^6\ \text{J}$

Section 1.3 Review

KEY QUESTIONS SOLUTIONS

1 The mercury is changing state from solid to liquid. It is melting; temperature does not change during phase transitions as the average kinetic energy does not change.

2 –39°C

3 357°C

4 $Q = mL_{fusion}$

$126 = 0.01 \times L_{fusion}$

$L_{fusion} = 126 \div 0.01 = 12\ 600$

$= 1.26 \times 10^4\ \text{J kg}^{-1}$

5 $Q = mL_{vapour}$

$3520 - 670 = 0.01 \times L_{vapour}$

$L_{vapour} = 2850 \div 0.01 = 285\ 000$

$= 2.85 \times 10^5\ \text{J kg}^{-1}$

6 $Q = mL_{fusion}$

$= 0.1 \times 22.5 \times 10^5$

$= 2.25 \times 10^5\ \text{J}$

7 Energy needed to raise the ice from –4.00°C to 0°C

$Q = mc\Delta T$

$= 0.100 \times 2100 \times 4.00$

$= 840\ \text{J}$

Energy needed to melt the ice at 0°C

$Q = mL_{fusion}$

$= 0.100 \times 3.34 \times 10^5$

$= 3.34 \times 10^4\ \text{J}$

Total energy $= 840 + (3.34 \times 10^4)$

$= 3.42 \times 10^4$

$= 34\ \text{kJ}$

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8 Hot water molecules have more energy than cold water molecules and will be able to leave the surface of the spa-pool water at a greater rate than cold water.

9 Most of the liquid has evaporated and the remaining liquid becomes colder as it does so, which in turn cools the floor.

Section 1.4 Heating and cooling

Worked example: Try yourself 1.4.1

CALCULATING THERMAL EQUILIBRIUM 1

4.00 kg of water initially at 85.0°C is mixed with 3.00 kg of water initially at 25.0°C. What is the final temperature of the water once thermal equilibrium is reached?	
Thinking	**Working**
Total energy lost by hot water = total energy gained by cold water That is, the energy change, ΔQ, is equal for the hot and cold water. Use $\Delta Q = mc\Delta T$ Assume no loss to the surrounding environment.	$\Delta Q_{hot} = \Delta Q_{cold}$ $m_{hot}c\Delta T_{hot} = m_{cold}c\Delta T_{cold}$
Since specific heat capacity of the water will be the same on both sides of the equation, the equation can be simplified.	$m_{hot}\Delta T_{hot} = m_{cold}\Delta T_{cold}$
Substitute the known values and simplify for the equilibrium temperature, T.	$4.00 \times (85.0 - T) = 3.00 \times (T - 25.0)$ $340 - 4.00T = 3.00T - 75.0$ $340 + 75.0 = 3.00T + 4.00T$ $415 = 7.00T$ $T = \frac{415}{7.00}$ $T = 59.3^{\circ}C$
Do a quick intuitive check. Does the answer make sense?	As most of water was warmer, the final temperature should be closer to the temperature of the original warmer water than to the original cooler water.

Worked example: Try yourself 1.4.2

CALCULATING THERMAL EQUILIBRIUM 2

A 75.0 g piece of copper is heated over a flame for several minutes. The copper is then plunged into an insulated, closed container containing 0.500 L of cool water, originally at 20.0°C. When thermal equilibrium is reached, the temperature of the water is found to be 22.0°C. If no water changes state to become steam and there are no other energy losses, then what was the temperature of the copper just before it was immersed in the water?	
Thinking	**Working**
Convert all masses to standard units (kg).	Mass of copper = 75.0 g = 0.0750 kg Mass of water = 0.500 kg (1.00 L of water = 1.00 kg)
Refer to Table 1.2.1 for the relevant specific heat capacity (c) values.	$c_{copper} = 390\ \mathrm{J\,kg^{-1}\,K^{-1}}$ $c_{water} = 4180\ \mathrm{J\,kg^{-1}\,K^{-1}}$
Total energy lost by copper = total energy gained by water That is, the energy change, ΔQ, is equal for the copper and the water.	$\Delta Q_{copper} = \Delta Q_{water}$ $m_c c_c \Delta T_c = m_w c_w \Delta T_w$
Substitute the known values, expand and simplify to solve for the initial temperature of the copper.	$m_c c_c \Delta T_c = m_w c_w \Delta T_w$ $0.0750 \times 390 \times (T_c - 22.0) = 0.500 \times 4180 \times (22.0 - 20.0)$ $29.25T_c - 643.5 = 4180$ $29.25T_c = 4823.5$ $T_{copper} = \frac{4823.5}{29.25}$ $T_{copper} = 165^\circ\mathrm{C}$

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Worked example: Try yourself 1.4.3

CHANGES OF STATE

Calculate the heat energy that must be lost, in J, to convert 5.00 kg of water vapour at 140.0°C into solid ice at 0.00°C.	
Thinking	**Working**
Identify the steps involved in the process	Step 1: Steam at 140.0°C to steam at 100.0°C Step 2: Steam at 100.0°C to water at 100.0°C Step 3: Water at 100.0°C to water at 0.00°C Step 4: Water at 0.00°C to ice at 0.00°C
Identify values for L and c for each step. Use tables 1.2.1, 1.3.1 and 1.3.2 to look up the values.	$c_{steam} = 2000\ \text{J kg}^{-1}\ \text{K}^{-1}$ $c_{water} = 4180\ \text{J kg}^{-1}\ \text{K}^{-1}$ $L_{fusion} = 3.34 \times 10^5\ \text{J kg}^{-1}$ $L_{vapour} = 22.5 \times 10^5\ \text{J kg}^{-1}$
Calculate the energy required for each step separately using the appropriate equation for specific heat or latent heat.	Step 1: Cooling the steam $Q_1 = mc\Delta T$ $= 5.00 \times 2000 \times 40.0$ $= 4.00 \times 10^5\ \text{J}$ Step 2: Condensing the steam $Q_2 = mL_{vapour}$ $= 5.00 \times 22.5 \times 10^5$ $= 1.125 \times 10^7\ \text{J}$ Step 3: Cooling the water $Q_3 = mc\Delta T$ $= 5.00 \times 4180 \times 100.0$ $= 2.09 \times 10^6\ \text{J}$ Step 4: Freezing the water $Q_4 = mL_{fusion}$ $= 5.00 \times 3.34 \times 10^5$ $= 1.67 \times 10^6\ \text{J}$
Add the energy required for each step together to find the total energy required.	$Q_T = Q_1 + Q_2 + Q_3 + Q_4$ $= (4.00 \times 10^5) + (1.125 \times 10^7) + (2.09 \times 10^6) + (1.67 \times 10^6)$ $= 1.54 \times 10^7\ \text{J}$

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Section 1.4 Review

KEY QUESTIONS SOLUTIONS

1 Aluminium. All other variables being the same, aluminium has the highest value for specific heat capacity so it will absorb the highest amount of thermal energy.

2 $T = 20.6°\text{C}$

$$Q_{\text{lost by hot}} = Q_{\text{gained by cold}}$$
$$m_h c\Delta T = m_c c\Delta T$$
$$m_h \Delta T = m_c \Delta T$$
$$10.0 \times (65.0 - T) = 80.0 \times (T - 15.0)$$
$$650.0 - 10.0T = 80.0T - 1200.0$$
$$90.0T = 1850.0$$
$$T = \frac{1850.0}{90.0}$$
$$= 20.56°\text{C}$$
$$= 20.6°\text{C (to 3 significant figures)}$$

3

$$Q_{\text{lost by copper}} = Q_{\text{gained by water}}$$
$$m_c c_c \Delta T = m_w c_w \Delta T$$
$$20.0 \times 390 \times (100.0 - T) = 5.00 \times 4180 \times (T - 20.0)$$
$$7.80 \times 10^5 - 7.80 \times 10^3 T = 2.09 \times 10^4 T - 4.18 \times 10^5$$
$$2.87 \times 10^4 T = 1.198 \times 10^6$$
$$T = \frac{1.198 \times 10^6}{2.87 \times 10^4 T}$$
$$= 41.7°\text{C}$$

4

$$Q_{\text{cooling}} = Q_{\text{warming}}$$
$$mc_{\text{water}}\Delta T = mc_{\text{water}}\Delta T$$
$$m_{\text{cooled}}\Delta T = m_{\text{heated}}\Delta T$$
$$m_{\text{cooled}} \times (45.0 - 36.0) = 12.0 \times (36.0 - 19.0)$$
$$m_{\text{cooled}} \times 9.0 = 204$$
$$m_{\text{cooled}} = \frac{204}{9.0}$$
$$= 22.7\text{ kg}$$

5

$$Q_{\text{lost by iron}} = Q_{\text{gain by water}}$$
$$m_i c_i \Delta T = m_w c_w \Delta T$$
$$598 \times 440 \times (1250 - T) = 938 \times 4180 \times (T - 21.0)$$
$$3.289 \times 10^8 - 2.631 \times 10^5 T = 3.9208 \times 10^6 T - 8.2338 \times 10^7$$
$$4.1839 \times 10^6 T = 4.1124 \times 10^8$$
$$T = \frac{4.1124 \times 10^8}{4.1839 \times 10^6}$$
$$= 98.3°\text{C}$$

6

$$Q_{\text{iron}} + Q_{\text{aluminium}} = Q_{\text{water}}$$
$$m_i c_i \Delta T + m_a c_a \Delta T = m_w c_w \Delta T$$
$$10.0 \times 440 \times (20.0 - T) + 10.0 \times 900 \times (20.0 - T) = 100 \times 4180 \times (T - 12.0)$$
$$8.80 \times 10^4 - 4.40 \times 10^3 T + 1.80 \times 10^5 - 9.00 \times 10^3 T = 4.18 \times 10^5 T - 5.016 \times 10^6$$
$$4.18 \times 10^5 T + 4.40 \times 10^3 T + 9.00 \times 10^3 T = 8.80 \times 10^4 + 1.80 \times 10^5 + 5.016 \times 10^6$$
$$4.314 \times 10^5 T = 5.284 \times 10^6$$
$$T = \frac{5.284 \times 10^6}{4.314 \times 10^5}$$
$$= 12.2°\text{C}$$

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7 $Q_{\text{total}} = Q_{\text{heating}} + Q_{\text{steam}}$

$$\begin{aligned} &= mc\Delta T + mL_v \\ &= 50.0 \times 4180 \times (100 - 20.0) + 50.0 \times 2.25 \times 10^6 \\ &= 1.672 \times 10^7 + 1.125 \times 10^8 \\ &= 1.29 \times 10^8 \text{ J} \end{aligned}$$

8

$$Q_{\text{steam}} + Q_{\text{water}} = Q_{\text{potatoes}}$$

$$m_s L_v + m_s c_w \Delta T = m_p c_p \Delta T$$

$$m_s \times 2.25 \times 10^6 + m_s \times 4180 \times (100 - 85.0) = 3.00 \times 3430 \times (85.0 - 12.5)$$

$$2.25 \times 10^6 m_s + 6.27 \times 10^4 m_s = 7.46 \times 10^5$$

$$2.3127 \times 10^6 m_s = 7.46 \times 10^5$$

$$\begin{aligned} m_s &= \frac{7.46 \times 10^5}{2.3127 \times 10^6} \\ &= 0.323 \text{ kg} \end{aligned}$$

9 $Q_{\text{total}} = mc\Delta T + mL_{\text{fusion}}$

$$\begin{aligned} &= 1.25 \times 233 \times (961 - 20.0) + 1.25 \times 1.11 \times 10^3 \\ &= 2.74 \times 10^5 + 1.39 \times 10^3 \\ &= 2.75 \times 10^5 \text{ J} \end{aligned}$$

10 $Q_{\text{total}} = mc_s\Delta T + mL_v + mc_w\Delta T$

$$\begin{aligned} &= 0.755 \times 2000 \times (110 - 100) + 0.755 \times 2.25 \times 10^6 + 0.755 \times 4180 \times (100 - 25.0) \\ &= 1.51 \times 10^4 + 1.70 \times 10^6 + 2.367 \times 10^5 \\ &= 1.95 \times 10^6 \text{ J} \end{aligned}$$

CHAPTER 1 REVIEW

1 A. The kinetic theory states that particles are in constant motion.

2 Temperature—the average kinetic energy of particles in a substance.

3 Heat refers to the energy that is transferred between objects, whereas temperature is a measure of the average kinetic energy of the particles within a substance.

4 **a** 5 + 273 = 278 K

b 200 – 273 = –73°C

5 The fixed points must be reproducible under any conditions. The starting point of the scale must be zero, with no negative values.

6 0°C is not the lowest value on the Celsius scale—negative values are possible. The freezing and boiling points of water are not fixed but vary with changing pressure.

7 As thermal equilibrium is reached, the balls must be at the same temperature.

8 B

9 The substance is changing state—in this case, it is melting. The heat energy is used to increase the potential energy of the particles in the solid instead of increasing their kinetic energy, so the temperature does not change. The energy needed to change from solid to liquid is the latent heat of fusion.

10 Both have the same kinetic energy as their temperatures are the same; however, the steam has more potential energy due to its change in state. Therefore the steam has greater internal energy.

11 The higher energy particles are escaping, leaving behind the lower energy particles. The result is that the average kinetic energy of the remaining particles decreases, thus the temperature drops.

12 $Q = mc\Delta T$

$$\begin{aligned} c &= \frac{Q}{m\Delta T} \\ &= \frac{5020}{2.00 \times 20} \\ &= 125.5 \text{ J kg}^{-1} \text{ K}^{-1} \\ &= 126 \text{ J kg}^{-1} \text{ K}^{-1} \end{aligned}$$

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13 $Q = mL$

$= 0.08 \times 0.88 \times 10^5$

$= 7.0\,\text{kJ}$

14 $c_{copper} = 390\,\text{J kg}^{-1}\,\text{K}^{-1}$, $c_{iron} = 440\,\text{J kg}^{-1}\,\text{K}^{-1}$. Copper requires less thermal energy to heat it than iron so will cool the water travelling through it less than iron. However, it is also a better conductor of heat so will require additional insulation to avoid transferring more heat to the surrounds.

15 $\Delta U = Q - W$

$= +14\,600 - (-2.65 \times 10^6)$

$= 2\,664\,600\,\text{J}$

$Q = mL_{fusion} + mc\Delta T$

$2\,664\,600 = 4.55 \times 3.34 \times 10^5 + 4.55 \times 4180 \times (T - 0)$

$T = 60°\text{C}$

16 Note that both the cup and the water must be cooled since there will be heat transfer between the two materials in contact.

$$Q_{\text{melting ice}} + Q_{\text{heating ice}} = Q_{\text{cooling water}} + Q_{\text{cooling cup}}$$

$$m_{\text{ice}}L_f + m_{\text{ice}}c_{\text{water}}\Delta T = m_{\text{water}}c_{\text{water}}\Delta T + m_{\text{copper}}c_{\text{copper}}\Delta T$$

$$m_{\text{ice}} \times 3.34\times10^5 + m_{\text{ice}} \times 4180\times(20-0) = 0.100\times4180\times(60-20) + 0.200\times390\times(60-20)$$

$$3.34\times10^5 m_{\text{ice}} + 8.36\times10^4 m_{\text{ice}} = 1.672\times10^4 + 3120$$

$$4.176\times10^5 m_{\text{ice}} = 1.984\times10^4$$

$$m_{\text{ice}} = \frac{1.984\times10^4}{4.176\times10^5}$$

$$= 4.75\times10^{-2}\ \text{kg}$$

17

$$Q_{\text{milk}} = Q_{\text{steam}} + Q_{\text{water}}$$

$$m_m c\Delta T = m_s L_f + m_w c_w \Delta T$$

$$0.425\times3930\times(70.0-4.00) = 2.25\times10^6 m + m\times4180\times(100-70.0)$$

$$1.10\times10^5 = 2.25\times10^6 m + 1.254\times10^5 m$$

$$2.375\times10^6 m = 1.10\times10^5$$

$$m = \frac{1.10\times10^5}{2.375\times10^6}$$

$$= 4.63\times10^{-2}\ \text{kg}$$

18

$$Q_{\text{lemon}} = Q_{\text{ice}} + Q_{\text{water}}$$

$$m_l c\Delta T = m_i L_f + m_w c_w \Delta T$$

$$0.468\times3850\times(20.0-3.00) = 3.34\times10^5 m + m\times4180\times(3.00-0.00)$$

$$3.063\times10^4 = 3.34\times10^5 m + 1.254\times10^4 m$$

$$3.063\times10^4 = 3.465\times10^5 m$$

$$m = \frac{3.063\times10^4}{3.465\times10^5}$$

$$= 8.84\times10^{-2}\ \text{kg}$$

19

$$Q_{\text{steam}} = Q_{\text{ice}}$$

$$m_s c_s \Delta T + m_s L_v + m_s c_w \Delta T = m_i c_i \Delta T + m_i L_f + m_i c_w \Delta T$$

$$m_s\times2000\times(115-100) + 2.25\times10^6 m_s + m_s\times4180\times(100-55.0) = 2.50\times2100\times(0-(-12.5)) + 2.50\times3.34\times10^5 + 2.50\times4180\times(55.0-0)$$

$$3.00\times10^4 m_s + 2.25\times10^6 m_s + 1.881\times10^5 m_s = 6.56\times10^4 + 8.35\times10^5 + 5.7475\times10^5$$

$$2.4681\times10^6 m_s = 1.4754\times10^6$$

$$m_s = \frac{1.4754\times10^6}{2.4681\times10^6}$$

$$= 0.598\ \text{kg}$$

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20

$$Q_{\text{iron}} = Q_{\text{water}}$$

$$m_i c_i \Delta T = m_w c_w \Delta T + m_w L_v$$

$$18.0 \times 440 \times (545 - T) = 1.50 \times 4180 \times (100 - 22.0) + 1.50 \times 2.25 \times 1$$

$$4.316 \times 10^6 - 7.920 \times 10^3 T = 4.8906 \times 10^5 + 3.375 \times 10^6$$

$$7.920 \times 10^3 T = 4.316 \times 10^6 - 3.8641 \times 10^6$$

$$T = \frac{4.519 \times 10^5}{7.920 \times 10^3}$$

$$= 57.1°\text{C}$$

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